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RENEWABLES

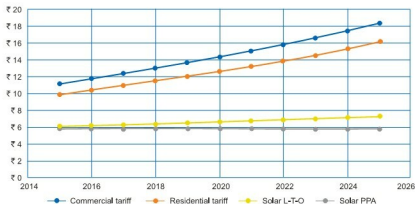


Fig. 5: Comparison of tariffs for different models

cialisation of solar energy increases as the size of the solar power system increases.

Also, cost per watt of energy from a solar system reduces considerably with reduction in cost per watt of the inverter. Panel cost reduces only marginally with an increase in system size. What does result in a drastic reduction in costs is drop in inverter costs. If you see Fig. 2, a considerable dip is seen in price per watt if the system size goes beyond 20kW, after which the price drops only marginally.

The investment

Let us now look at a representative cash flow of a solar power system for a typical 40kW system. If the initial investment is about ₹ 2,71,000, the benefits that you can expect include the facility of net-metering, giving you a return of about 20 per cent annually for 20 years, depending upon the per unit cost of the energy generated and the load used. To understand this better, please see the example given in the box on first page of this article.

Fig. 4 clearly shows that the money gets recovered in only four years, beyond which electricity is free. The quantum that can be saved (cumulatively) is as high as ₹ 2 million.

This cash flow was calculated (Fig. 5) assuming that grid tariff

will rise at a conservative five per cent per annum. Historically, it has been higher than that. Solar tariffs would either rise at two per cent per annum in PPA mode or remain the same in loan mode, in which 30 per cent cost of the system is paid upfront.

You can comfortably say that solar makes great sense at this point and can result in high savings. It is not the cost of installation that has been holding back end customers from going solar, it is the high cost of trying something for which there is lack of demonstration available.

Today, there are multiple sites in New Delhi (and elsewhere) where grid-connected solar PV systems have become a reality. National Productivity Council is one such site.

Also, since standardisation has not been seen in the industry yet, an aggregator (like Flipkart, or SolarCity in the USA) might do the magic in enabling massive uptake. Sunkalp Energy has shifted to the aggregator model (through online portal) and Ministry of New and Renewable Energy also seems to be following this approach.

So, when installation of solar becomes more common, end customers will be able to install their own systems. All they would need to do is, select their components and services online and have the solar system set up in no time. **EFY**